

Smart Air Quality Analysis and Prediction using Linear Regression

Project Overview

This project analyzes air quality data from major Indian cities and predicts PM2.5 concentration using Linear Regression. The system also performs pollution hotspot ranking, seasonal analysis, health risk assessment, and generates air quality alerts.

Objectives

- Analyze air quality data from major Indian cities.
- Identify pollution trends and hotspot cities.
- Study seasonal variations in air pollution.
- Predict PM2.5 concentration using Linear Regression.
- Provide health risk assessment and air quality alerts.

Dataset

- Dataset Size: 29,531 records
- Cities Analyzed: Delhi, Mumbai, Kolkata, Chennai, Hyderabad, Bengaluru
- Features Used: PM2.5, PM10, CO, NO₂, SO₂, O₃

Key Insights

Pollution Hotspot Ranking

Delhi recorded the highest pollution score among all analyzed cities, indicating consistently higher pollution levels compared to other cities.

Seasonal Analysis

Winter showed the highest average PM2.5 concentration, while Monsoon recorded the lowest pollution levels.

Most Polluted Day

The highest PM2.5 concentration observed in the dataset was 299.85 in Delhi.

AQI Analysis

Most observations belonged to the Good and Satisfactory AQI categories, while a smaller portion fell into Poor AQI conditions.

Model Performance

- R^2 Score: 0.755
- MAE: 14.01
- MSE: 515.29

The Linear Regression model successfully explained approximately 75.5% of the variation in PM2.5 concentrations.

Features Implemented

- AQI Classification
- Health Risk Assessment
- Pollution Hotspot Ranking
- Most Polluted Day Detection
- Season-wise Pollution Analysis
- PM2.5 Prediction
- Air Quality Alert System
- Health Recommendation Engine

Conclusion

The project successfully analyzes air pollution patterns and predicts PM2.5 concentration using Linear Regression. The findings indicate that Delhi experiences the highest pollution levels and that winter is the most polluted season. The developed system can assist users by providing pollution alerts and health recommendations based on predicted air quality levels.